6.4 Transformations of Exponential and Logarithmic Functions

Name _____ Period ___ Date _____

Algebra II 6.4 · Student sheet

I can…

- describe transformations of exponential and logarithmic functions.
- find the asymptote and domain of a transformed function.
- write a rule for a transformed exponential or logarithmic function.

Vocabulary · fill in the missing word

Exponential transformations — $g(x) = a \cdot b^{x-h} + k$: _____ asymptote $y = k$

Logarithmic transformations — $g(x) = a \cdot \log_b(x - h) + k$: vertical _____ $x = h$, domain $x > h$

Part 1 · Desmos sliders

In Desmos, type $y = a \cdot 2^{x-h} + k$ with sliders. Then type $y = a \cdot \log_2(x - h) + k$. Record what happens.

Slider	What it does to $y = a \cdot 2^{x-h} + k$	What it does to $y = a \cdot \log_2(x - h) + k$
h		
k		
a negative		
The asymptote is		

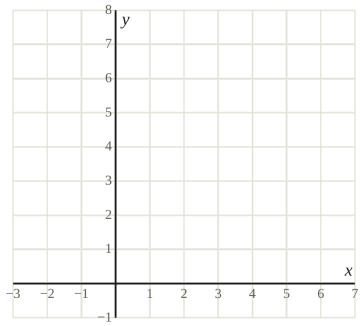
Part 2 · Notes and examples (fill in each note, then work the example)

Exponential transformations: For $g(x) = a \cdot b^{x-h} + k$, the horizontal asymptote is _____.

Translations: h moves the graph _____ or _____; k moves it _____ or _____.

Example 1

Describe the transformation of $f(x) = 4^x$ represented by $g(x) = 4^{x-3} + 1$. Give the asymptote of g.



Reflections: $a < 0$ reflects the graph in the _____; a negative x reflects it in the _____.

Example 2

Describe the transformation of $f(x) = (\frac{1}{2})^x$ represented by $g(x) = -(\frac{1}{2})^x - 3$. Give the asymptote of g.

Logarithmic transformations: For $g(x) = a \cdot \log_b(x - h) + k$, the vertical asymptote is _____ and the domain is _____.

Example 3

Describe the transformation of $f(x) = \log_2 x$ represented by $g(x) = \log_2(x + 4) - 2$. Give the asymptote and domain of g.

Example 4

Describe the transformation of $f(x) = \log x$ represented by $g(x) = \frac{1}{2} \log x + 1$.

Example 5

Let g be a vertical stretch by a factor of 2, followed by a translation 3 units left, of $f(x) = e^x$. Write a rule for g.

Example 6

Let g be a reflection in the y-axis, followed by a translation 1 unit up, of $f(x) = \log_3 x$. Write a rule for g.

Part 3 · Partner practice

Partner 1

Describe the transformation of $f(x) = 3^x$ represented by $g(x) = 3^{x+2} - 5$. Give the asymptote of g.

Partner 2

Let g be a translation 6 units right, followed by a reflection in the x-axis, of $f(x) = \log x$. Write a rule for g.